

Section 5.1 If  $f: \mathbb{R} \rightarrow \mathbb{R}$  satisfies that

$$|f(x) - f(y)| \leq (x - y)^2 \text{ for all } x, y \in \mathbb{R}$$

Then  $f$  is constant.

Proof. Since  $\left| \frac{f(x) - f(y)}{x - y} \right| \leq |x - y|$  for all  $x, y \in \mathbb{R}$ ,

Given  $x \in \mathbb{R}$ ,  $f'(x)$  exists with  $f'(x) = 0$ .

Now, by the Mean Value Theorem, for any  $a < b$ , there exists  $c \in (a, b)$  s.t.

$$f(a) - f(b) = (a - b) \cdot f'(c) = 0$$

Thus  $f$  is constant.

Section 5.2. Let  $f: (a, b) \rightarrow \mathbb{R}$  be given. If  $f'(x) > 0$  for all  $x \in (a, b)$ , then  $f$  is injective.

Moreover, the inverse  $f^{-1}: f[(a, b)] \rightarrow (a, b)$  exists and is differentiable with

$$(f^{-1})'(f(x)) = \frac{1}{f'(x)} \quad (a < x < b)$$

Proof. Let  $a < x < y < b$ . Then, by the Mean Value Theorem, there exists  $c \in (x, y)$  s.t.

$$f(y) - f(x) = f'(c) \cdot (y - x) > 0$$

Therefore,  $f$  is strictly increasing, thus injective. Now, the restriction

$$f: (a, b) \rightarrow f[(a, b)]$$

is bijective, thus there exists an inverse

$$f^{-1}: f[(a, b)] \rightarrow (a, b): f(x) \mapsto x$$

Now,  $(f^{-1} \circ f)(x) = x$  implies that

Section 5.3. Let  $g: \mathbb{R} \rightarrow \mathbb{R}$  be given with  $|g'| \leq M$  for some  $M > 0$ .

Fix  $\varepsilon > 0$ , and define  $f(x) = x + \varepsilon \cdot g(x)$ . prove that  $f$  is injective if  $\varepsilon$  is small enough.

Proof. If  $\varepsilon < \frac{1}{M}$ , then  $f'(x) = 1 + \varepsilon \cdot g'(x) > 0$ . Therefore  $f$  is injective.

Section 5.4. If  $C_0, \dots, C_n \in \mathbb{R}$  satisfies

$$C_0 + \frac{C_1}{2} + \frac{C_2}{3} + \dots + \frac{C_{n-1}}{n} + \frac{C_n}{n+1} = 0$$

, then the polynomial

$$C_0 + C_1 x + C_2 x^2 + \dots + C_{n-1} x^{n-1} + C_n x^n = p(x)$$

has a real root in  $(0, 1)$ .

Proof. Define  $g(x) = C_0 x + \frac{C_1}{2} x^2 + \dots + \frac{C_{n-1}}{n} x^n + \frac{C_n}{n+1} x^{n+1}$

Then,  $g(0) = 0$  clearly, and  $g(1) = 0$  by the assumption.

By the Mean-Value Theorem, there exists  $\alpha \in (0, 1)$  s.t.

$$0 = g(1) - g(0) = g'(\alpha) \cdot (1 - 0) = g'(\alpha)$$

Directly,  $p(x) = g'(x)$ . therefore  $g'(\alpha) = p(\alpha) = 0$ .

Section 5.5 Let  $f: (0, \infty) \rightarrow \mathbb{R}$  be differentiable.

If  $f(x) \rightarrow 0$  as  $x \rightarrow \infty$ , then  $f(x+1) - f(x) \rightarrow 0$  as  $x \rightarrow \infty$ .

Proof. By the MVT, for any  $x > 0$ , there exists  $c_x \in (x, x+1)$  s.t.

$$f(x+1) - f(x) = f'(c_x) \cdot (1 - 0) = f'(c_x).$$

Let  $\varepsilon > 0$  be given. Then, there exists  $\delta > 0$  s.t.  $x \geq \delta \Rightarrow |f(x)| < \varepsilon$ .

For this  $\delta > 0$ , for any  $x \geq \delta$ , there is  $c_x \in (x, x+1)$  s.t.

$$|f(x+1) - f(x)| = |f'(c_x)| < \varepsilon$$

Thus proved.

Section 5.6. Suppose that  $f: (0, \infty) \rightarrow \mathbb{R}$  satisfies that:

- 1)  $f$  is continuous for  $x \geq 0$
- 2)  $f'$  exists for  $x > 0$
- 3)  $f'$  increases monotonically.
- 4)  $f(0) = 0$ .

Then,  $g: (0, \infty) \rightarrow \mathbb{R}; x \mapsto \frac{f(x)}{x}$  increases monotonically.

Proof. Since  $g(x) = \frac{x f(x) - f(x^2)}{x^2}$ , enough to show  $f(x) \geq \frac{f(y)}{x} = g(y)$ .

observe that  $g(x) = \frac{f(x)}{x} = \frac{f(x) - f(0)}{x - 0} = f'(c_x)$  for some  $0 < c_x < x$ , by the MVT.

Now, since  $f'$  increases monotonically,  $f'(x) \geq g(x) = f'(c_x)$ .

Section 5.7. Let  $f, g: \mathbb{R} \rightarrow \mathbb{R}$  be given.

Suppose that  $f', g'$  exist and  $g'(x) \neq 0$  for all  $x \in \mathbb{R}$ , and  $f(x) = g(x) = 0$ .

Prove  $\lim_{t \rightarrow x} \frac{f(t)}{g(t)} = \frac{f'(x)}{g'(x)}$

Proof. Since  $f', g'$  exist, and  $f(0) = g(0) = 0$ ,

$$\lim_{t \rightarrow x} \frac{f(t)}{g(t)} = \lim_{t \rightarrow x} \frac{\frac{f(t) - f(x)}{t - x}}{\frac{g(t) - g(x)}{t - x}} = \frac{f'(x)}{g'(x)}$$

Section 5.8.  $\sum_{n=1}^{\infty} \sin\left(\frac{1}{n}\right)$  converges?

$\frac{\sin \frac{1}{n}}{\frac{1}{n}} \rightarrow 1$  but  $\sum \frac{1}{n}$  diverges, thus

Section 5.8. Let  $f: [a, b] \rightarrow \mathbb{R}$  be given. Suppose  $f'$  exists and is continuous on  $[a, b]$ .

Given  $\varepsilon > 0$ , there exists  $\delta > 0$  s.t.  $\forall x, t \in [a, b], |x - t| < \delta \Rightarrow \left| \frac{f(x) - f(t)}{x - t} - f'(x) \right| < \varepsilon$

Proof. Since  $f'$  is continuous on compact, it is continuous uniformly.

Given  $\varepsilon > 0$ , there exists  $\delta > 0$  s.t.  $\forall x, t \in [a, b],$

$$|x - t| < \delta \Rightarrow |f'(x) - f'(t)| < \varepsilon$$

Meanwhile, by the Mean-Value Thm,

$$f'(s) = \frac{f(x) - f(t)}{x - t} \text{ for some } s \text{ between } x \text{ and } t.$$

$$\text{Now, } |x - t| < \delta \Rightarrow \left| \frac{f(x) - f(t)}{x - t} - f'(t) \right| = |f'(s) - f'(t)| < \varepsilon$$

because  $|s - t| \leq |x - t|$ .

[Section 5.9.

$$f: \mathbb{R} \rightarrow \mathbb{R}; x \mapsto \begin{cases} x+3 & x \geq 0 \\ 3 & x < 0 \end{cases}$$

[Section 5.10. Let  $f, g: (0,1) \rightarrow \mathbb{C}$  be differentiable such that

$$\lim_{x \rightarrow 0} f(x) = \lim_{x \rightarrow 0} g(x) = 0 \quad \text{and} \quad \lim_{x \rightarrow 0} f'(x), \lim_{x \rightarrow 0} g'(x) (\neq 0) \quad \text{exist. Then,}$$

$$\lim_{x \rightarrow 0} \frac{f(x)}{g(x)} = \lim_{x \rightarrow 0} \frac{f'(x)}{g'(x)}$$

Proof. Observe that  $\frac{f(x)}{g(x)} = \left[ \frac{f(x)}{x} - \lim_{t \rightarrow 0} \frac{f(t)}{t} \right] \cdot \frac{x}{g(x)} + \lim_{t \rightarrow 0} \frac{f(t)}{t} \cdot \frac{x}{g(x)}$

Meanwhile, by the L'Hopital Law,

$$\lim_{x \rightarrow 0} \frac{f(x)}{x} = \lim_{x \rightarrow 0} \left[ \frac{\operatorname{Re} f(x)}{x} + i \cdot \frac{\operatorname{Im} f(x)}{x} \right]$$

$$= \lim_{x \rightarrow 0} \left[ \operatorname{Re} f'(x) + i \cdot \operatorname{Im} f'(x) \right] = \lim_{x \rightarrow 0} f'(x)$$

Similarly  $\lim_{x \rightarrow 0} \frac{x}{g(x)} = \left( \lim_{x \rightarrow 0} g'(x) \right)^{-1}$ . Now,

$$\lim_{x \rightarrow 0} \frac{f(x)}{g(x)} = \left( \lim_{x \rightarrow 0} f'(x) \right) \cdot \left( \lim_{x \rightarrow 0} g'(x) \right)^{-1}$$

# Section 5.11.

Suppose that  $f$  is defined on a neighborhood of  $x \in \mathbb{R}$ , and  $f''(x)$  exists. Prove that

$$\lim_{h \rightarrow 0} \frac{f(x+h) + f(x-h) - 2f(x)}{h^2} = f''(x)$$

Proof. By the L'Hopital Law,

$$\lim_{h \rightarrow 0} \frac{f(x+h) + f(x-h) - 2f(x)}{h^2} \stackrel{①}{=} \lim_{h \rightarrow 0} \frac{f'(x+h) - f'(x-h)}{2h} \stackrel{②}{=} \frac{1}{2} \cdot \lim_{h \rightarrow 0} (f''(x+h) + f''(x-h)) = f''(x)$$

But, if  $f''(x)$  does not exist, then we can not apply L'Hopital Law in ②.

Define  $f: \mathbb{R} \rightarrow \mathbb{R}; x \mapsto \begin{cases} 0 & \text{if } x < 0 \\ x^2 & \text{if } x \geq 0 \end{cases}$

Then,  $f''(0)$  does not exist, But

$$\lim_{h \rightarrow 0} \frac{h^2 + 0 - 0}{h^2} = 1$$

exists.

## For all Try

By the Mean-Value Theorem, for any  $h > 0$ ,

$f(x+h) - f(x) = f'(t_1)h$  and  $f(x-h) - f(x) = f'(t_2)h$  for some  $t_1 \in (x, x+h)$ ,  $t_2 \in (x-h, x)$

$$\frac{f'(t_1) + f'(t_2)}{h} = 2f''(x)$$

Section 5.13, Let  $\alpha \in \mathbb{R}$ ,  $c > 0$ . Define

$$f: [-1, 1] \rightarrow \mathbb{R}; x \mapsto \begin{cases} 0 & \text{if } x=0 \\ x^\alpha \cdot \sin\left(\frac{1}{|x|^c}\right) & \text{if } x \neq 0 \end{cases}$$

| Equivalence Condition Table | $f$             | $f'$              | $f''$              |
|-----------------------------|-----------------|-------------------|--------------------|
| Continuous                  | $\alpha > 0$    | $\alpha > 1+c$    | $\alpha > 2+2c$    |
| bounded                     | $\alpha \geq 0$ | $\alpha \geq 1+c$ | $\alpha \geq 2+2c$ |
| Exists at $x=0$             | Always          | $\alpha > 1$      | $\alpha > 2+c$     |

Proof. Suppose that  $\alpha > 0$ . Given  $\epsilon > 0$ , put  $\delta = \sqrt[\alpha]{\epsilon}$ . Then,

$$|x| < \delta \Rightarrow |x^\alpha \cdot \sin\left(\frac{1}{|x|^c}\right)| \leq |x|^\alpha < \epsilon.$$

Thus continuous. And, suppose that  $\alpha \leq 0$ . Then, a sequence

$$x_n = \sqrt[c]{\frac{1}{(2n+\frac{1}{2})\pi}}$$

converges to 0, but

$$f(x_n) = \left(\frac{1}{(2n+\frac{1}{2})\pi}\right)^{\frac{\alpha}{c}} \cdot \underbrace{\sin\left(2n\pi + \frac{\pi}{2}\right)}_{=1}$$

not converges to  $f(0)=0$ . Except  $x \neq 0$ , clearly continuous.

$f'(0)$  exists iff

Suppose  $\alpha \geq 1+c$ . Then, for  $x \neq 0$ ,

$$\lim_{t \rightarrow 0} \frac{f(t) - f(0)}{t - 0} = \lim_{t \rightarrow 0} \left( t^{\alpha-1} \cdot \sin\left(\frac{1}{|t|^c}\right) \right)$$

exists. Since it equals to  $t^{-1} \cdot f(t)$ , the limit exists only  $\alpha-1 > 0$ .

observe that: for  $x > 0$ ,

$$\begin{aligned} f'(x) &= \alpha x^{\alpha-1} \sin\left(\frac{1}{x^c}\right) + x^\alpha \cdot (-c) \cdot x^{-c-1} \cdot \cos\left(\frac{1}{x^c}\right) \\ &= \alpha x^{\alpha-1} \sin\frac{1}{x^c} - c \cdot x^{\alpha-(1+c)} \cdot \cos\frac{1}{x^c} \end{aligned}$$

is bounded iff  $\alpha-(1+c) \geq 0$ .

d).  $f'$  is continuous at  $x=0$  iff  $f'(0) = \lim_{x \rightarrow 0} f'(x)$  (clearly, existence of  $f'(0)$  is assumed).  
If  $\alpha > 1+c$ , then  $f(x) \rightarrow 0$  as  $x \rightarrow 0$ , thus

$$f'(0) = \lim_{x \rightarrow 0} \frac{f(x)}{x} = \lim_{x \rightarrow 0} f'(x)$$

by the L'Hopital Law. And, suppose  $\alpha \leq 1+c$ . Then,

$$f'(x) = \alpha x^{\alpha-1} \sin \frac{1}{x^c} - c \cdot x^{\alpha-(1+c)} \cdot \cos \frac{1}{x^c}$$

does not converge as  $x \rightarrow 0$ .

Section 5.14. Let  $f: (a,b) \rightarrow \mathbb{R}$  be differentiable.

$f$  is convex if and only if  $f'$  is monotonically increasing.



Section 5.15. Let  $f: (a, \infty) \rightarrow \mathbb{R}$  be twice-differentiable.

Put  $M_k = \sup_{x \in (a, \infty)} |f^{(k)}(x)|$ ,  $k=0, 1, 2$ . Then  $M_1^2 \leq 4M_0M_2$ .

Proof. Given  $x > a$  and  $h > 0$ ,

$$f(x+2h) = f(x) + f'(x) \cdot 2h + \underbrace{f''(t) \cdot \frac{2h^2}{2!}}_{= \frac{2h^2}{2!}} \quad \text{for some } t \in (x, x+2h)$$

$$\text{Therefore } |f'(x)| \leq \left| \frac{f(x+2h) - f(x)}{2h} - h \cdot f''(t) \right| \leq \frac{M_0}{h} + M_2 h$$

Take  $h = \sqrt{\frac{M_0}{M_2}}$ . Then,

$$\forall x > a, |f'(x)| \leq \sqrt{M_0 M_2} + \sqrt{M_0 M_2} = 2\sqrt{M_0 M_2}$$

which implies that  $M_1 \leq 2\sqrt{M_0 M_2}$ .

Section 5.16. Let  $f: (0, \infty) \rightarrow \mathbb{R}$  be twice-differentiable,

$f''$  is bounded,

$f(x) \rightarrow 0$  as  $x \rightarrow \infty$ .

Then,  $f'(x) \rightarrow 0$  as  $x \rightarrow \infty$ .

Proof. Since for any  $a > 0$ ,

$$0 \leq \sup_{x \in (a, \infty)} |f'(x)| \leq 2 \sqrt{\sup_{x \in (a, \infty)} |f(x)| \cdot M} \quad (\text{where } M \text{ is bound for } |f''|)$$

Letting  $a \rightarrow \infty$  we obtain the result.

Section 5.17. Let  $f: [-1, 1] \rightarrow \mathbb{R}$  be three-times differentiable s.t.  
 $f(-1)=0$ ,  $f(0)=0$ ,  $f(1)=1$ ,  $f'(0)=0$ .

Then,  $f^{(3)}(x) \geq 3$  for some  $x \in (-1, 1)$ .

Proof. By the Taylor's Theorem,

$$f(1) = f(0) + f'(0) + \frac{f''(0)}{2!} + \frac{f^{(3)}(s)}{3!} \quad \text{for some } s \in (0, 1)$$

$$\text{and } f(-1) = f(0) - f'(0) + \frac{f''(0)}{2!} - \frac{f^{(3)}(t)}{3!} \quad \text{for some } t \in (-1, 0)$$

Combining above, then

$$f(1) - f(-1) = 2f'(0) + \frac{1}{3!} (f^{(3)}(s) + f^{(3)}(t))$$

Thus  $f^{(3)}(s) + f^{(3)}(t) = 6$ , therefore either one bigger than 3.

Q. Why  $f'(0)=0$  is needed?

Section 5.18.

Let  $f: [a, b] \rightarrow \mathbb{R}$ ,  $n \in \mathbb{N}$ ,  $\alpha, \beta \in [a, b]$ . Suppose  $f^{(n-1)}$  exists on  $[a, b]$ . Define

$$Q(t) = \frac{f(t) - f(\beta)}{t - \beta} \quad (t \neq \beta).$$

$$\text{Then } f(\beta) = \sum_{k=0}^{n-1} \frac{f^{(k)}(\alpha)}{k!} \cdot (\beta - \alpha)^k + \frac{Q^{(n-1)}(\alpha)}{(n-1)!} \cdot (\beta - \alpha)^n$$

Proof. Derivating  $(n-1)$  times to:

$$(t - \beta) \cdot Q(t) = f(t) - f(\beta)$$

Then,

$$Q(t) + (t - \beta) Q'(t) = f'(t)$$

$$Q'(t) + Q''(t) + (t - \beta) Q'''(t) = f''(t)$$

;

$$(n-1) \cdot Q^{(n-2)}(t) + (t - \beta) \cdot Q^{(n-1)}(t) = f^{(n-1)}(t)$$

$$\text{Now, } f(\beta) = f(t) + Q(t) \cdot (\beta - t)$$

$$= f(t) + f'(t) \cdot (\beta - t) + Q'(t) (\beta - t)^2$$

$$= f(t) + f'(t) (\beta - t) + \frac{f''(t)}{2!} (\beta - t)^2 + \frac{Q''(t)}{2!} (\beta - t)^3$$

$$= \sum_{k=0}^{n-1} \frac{f^{(k)}(t)}{k!} \cdot (\beta - t)^k + \frac{Q^{(n-1)}(t)}{(n-1)!} \cdot (\beta - t)^n.$$

Section 5. 1a. Let  $f: (-1, 1) \rightarrow \mathbb{R}$  be a function such that  $f'(0)$  exists.

Suppose  $-1 < \alpha_n < \beta_n < 1$  and  $\lim_{n \rightarrow \infty} \alpha_n = \lim_{n \rightarrow \infty} \beta_n = 0$ . Define the difference quotient:

$$D_n = \frac{f(\beta_n) - f(\alpha_n)}{\beta_n - \alpha_n}$$

1). If  $\alpha_n < 0 < \beta_n$ , then  $\lim_{n \rightarrow \infty} D_n = f'(0)$

2). If  $0 < \alpha_n < \beta_n$  and  $\left\{ \frac{\beta_n}{\beta_n - \alpha_n} \right\}$  is bounded, then  $\lim_{n \rightarrow \infty} D_n = f'(0)$

3). If  $f'$  is continuous in  $(-1, 1)$ , then  $\lim_{n \rightarrow \infty} D_n = f'(0)$

4). Find Counter examples.

Proof. Observe that

$$\begin{aligned} \left| \frac{f(\beta_n) - f(\alpha_n)}{\beta_n - \alpha_n} - f'(0) \right| &= \left| \frac{f(\beta_n) - f(0)}{\beta_n - \alpha_n} - \frac{f(\alpha_n) - f(0)}{\beta_n - \alpha_n} - f'(0) \right| \\ &= \left| \frac{f(\beta_n) - f(0)}{\beta_n - 0} \cdot \frac{\beta_n}{\beta_n - \alpha_n} - \frac{f(\alpha_n) - f(0)}{\alpha_n - 0} \cdot \frac{\alpha_n}{\beta_n - \alpha_n} - f'(0) \right| \\ &\leq \left| \frac{\beta_n}{\beta_n - \alpha_n} \right| \cdot \left| \frac{f(\beta_n) - f(0)}{\beta_n - 0} - f'(0) \right| + \left| \frac{\alpha_n}{\beta_n - \alpha_n} \right| \cdot \left| \frac{f(\alpha_n) - f(0)}{\alpha_n - 0} - f'(0) \right| \quad (*) \end{aligned}$$

Now, if  $\left\{ \frac{\beta_n}{\beta_n - \alpha_n} \right\}$  is bounded  $(*)$  tends to 0 as  $n \rightarrow \infty$ .  $\left( \left| \frac{\alpha_n}{\beta_n - \alpha_n} \right| \leq \left| \frac{\beta_n}{\beta_n - \alpha_n} \right| \right)$

If  $\alpha_n < 0 < \beta_n$ ,  $\left| \frac{\beta_n}{\beta_n - \alpha_n} \right| \leq \left| \frac{\beta_n}{\beta_n} \right| = 1$ . Thus 1), 2) are proved.

And 3) can be solved by applying MVT. For any  $n \in \mathbb{N}$ , there exists  $c_n \in (\alpha_n, \beta_n)$  s.t.

$$D_n = \frac{f(\beta_n) - f(\alpha_n)}{\beta_n - \alpha_n} = f'(c_n)$$

Now,  $c_n \rightarrow 0$  as  $n \rightarrow \infty$  and  $f'$  is continuous,

$$D_n = f'(c_n) \rightarrow f'(0) \text{ as } n \rightarrow \infty.$$

Try for 18.

$$\begin{aligned}
 &= \left| \frac{\beta_n}{\beta_n - \alpha_n} \right| \cdot \left| \frac{f(\beta_n) - f(\alpha)}{\beta_n - \alpha} - \frac{f(\alpha_n) - f(\alpha)}{\beta_n} - \frac{(\beta_n - \alpha_n) \cdot f'(\alpha)}{\beta_n} \right| \\
 &= \left| \frac{\beta_n}{\beta_n - \alpha_n} \right| \cdot \left| \frac{f(\beta_n) - f(\alpha)}{\beta_n - \alpha} - f'(\alpha) - \left( \frac{f(\alpha_n) - f(\alpha)}{\beta_n} - \frac{\alpha_n \cdot f'(\alpha)}{\beta_n} \right) \right| \\
 &\leq \left| \frac{\beta_n}{\beta_n - \alpha_n} \right| \cdot \left| \frac{f(\beta_n) - f(\alpha)}{\beta_n - \alpha} - f'(\alpha) \right| + \left| \frac{\alpha_n}{\beta_n - \alpha_n} \right| \cdot \left| \frac{f(\alpha_n) - f(\alpha)}{\alpha_n - \alpha} - f'(\alpha) \right|
 \end{aligned}$$

$$\left| \frac{f(\beta_n) - f(\alpha)}{\beta_n - \alpha_n} \right| = \left| \frac{\beta_n}{\beta_n - \alpha_n} \cdot \frac{f(\beta_n) - f(\alpha)}{\beta_n - \alpha} \right| \leq M \cdot \left| \frac{f(\beta_n) - f(\alpha)}{\beta_n - \alpha} \right| \rightarrow M \cdot |f'(\alpha)|.$$

Given  $\epsilon > 0$ , there exists  $\delta > 0$  s.t.  $0 < |x| < \delta \Rightarrow \left| \frac{f(x) - f(\alpha)}{x - \alpha} - f'(\alpha) \right| < \epsilon$ .  
 , since  $f'(\alpha)$  exists. Take  $N \in \mathbb{N}$  s.t.

$$n \geq N \Rightarrow -\frac{\delta}{2} < \alpha_n < 0 \text{ and } 0 < \beta_n < \frac{\delta}{2}$$

(Note: 여기서 이상함을 느끼게 되는데, 좌변 형태와 다른 식들은 불확실 어떻게 할까?)

Meanwhile, since  $f'(\alpha)$  exists,  $f$  is continuous at  $\alpha$ . Therefore,

$$\lim_{n \rightarrow \infty} f(\alpha_n) = \lim_{n \rightarrow \infty} f(\beta_n) = f(\alpha)$$

$$\begin{aligned}
 \left| \frac{f(\beta_n) - f(\alpha_n)}{\beta_n - \alpha_n} - f'(\alpha) \right| &= \left| \frac{f(\beta_n) - f(\alpha)}{\beta_n - \alpha_n} - \frac{f(\alpha_n) - f(\alpha)}{\beta_n - \alpha_n} - f'(\alpha) \right| \\
 &= \left| \frac{\beta_n}{\beta_n - \alpha_n} \right| \cdot \left| \frac{f(\beta_n) - f(\alpha)}{\beta_n - \alpha} - \frac{f(\alpha_n) - f(\alpha)}{\beta_n} - \frac{(\beta_n - \alpha_n) \cdot f'(\alpha)}{\beta_n} \right| \\
 &= \left| \frac{\beta_n}{\beta_n - \alpha_n} \right| \cdot \left| \frac{f(\beta_n) - f(\alpha)}{\beta_n - \alpha} - f'(\alpha) - \left( \frac{f(\alpha_n) - f(\alpha)}{\beta_n} - \frac{\alpha_n \cdot f'(\alpha)}{\beta_n} \right) \right|
 \end{aligned}$$

$$\left| \frac{f(\beta_n) - f(\alpha)}{\beta_n - \alpha_n} \right| = \left| \frac{\beta_n}{\beta_n - \alpha_n} \cdot \frac{f(\beta_n) - f(\alpha)}{\beta_n - \alpha} \right| \leq M \cdot \left| \frac{f(\beta_n) - f(\alpha)}{\beta_n - \alpha} \right| \rightarrow M \cdot |f'(\alpha)|.$$

Section 5.22. Let  $f: \mathbb{R} \rightarrow \mathbb{R}$  be a differentiable s.t.  $f'(t) \neq 1$  for all  $t \in \mathbb{R}$ .

- 1).  $f$  has at most one fixed point.
- 2).  $f(t) = t + (1 + e^t)^{-1}$  has no fixed point, although  $0 < f'(t) < 1$  for all  $t \in \mathbb{R}$ .
- 3). If  $\sup_{t \in \mathbb{R}} |f'(t)| < 1$ , then  $f$  has a fixed point.

Proof. Let  $\alpha, \beta$  be two distinct fixed points. Then, by the Mean-value Theorem,

$$\frac{f(\beta) - f(\alpha)}{\beta - \alpha} = 1 = f'(c) \quad \text{for some } c.$$

It is contradiction. 2) is trivial.  $f(t) - t = \frac{1}{1 + e^t} \neq 0$ . But

$$f'(t) = 1 - (1 + e^t)^{-2} \cdot e^t = 1 - \frac{e^t}{(1 + e^t)^2}$$

is contained in  $(0, 1)$ . Let's show 3), The Banach-fixed point Theorem.

Solve in next page

Problem. Let  $f: I \rightarrow I$  be a differentiable with  $\sup |f'(x)| < 1$ .

Then  $f$  has a fixed point.

Proof. Construct a sequence. Let  $x_1 = \alpha \in I$ ,  $x_{n+1} = f(x_n)$   $n \geq 1$ .

Take  $\beta = \sup |f'(x)| < 1$ . Given  $n \in \mathbb{N}$ , if  $x_n = x_{n-1}$  then  $x_n$  is the fixed point. If  $x_n \neq x_{n-1}$ ,

$$|x_{n+1} - x_n| = |f(x_n) - f(x_{n-1})| = |x_n - x_{n-1}| \cdot |f'(c)| \leq \beta \cdot |x_n - x_{n-1}| \quad \text{by the MVT.}$$

$$\begin{aligned} \text{Now, } |x_{n+1} - x_n| &= |f(x_n) - f(x_{n-1})| \leq \beta |x_n - x_{n-1}| \\ &= \beta \cdot |f(x_{n-1}) - f(x_{n-2})| \leq \beta^2 |x_{n-1} - x_{n-2}| \\ &\vdots \\ &\leq \beta^{n-1} |x_2 - x_1|. \end{aligned}$$

Given  $\epsilon > 0$ , put  $N \in \mathbb{N}$  such that  $\beta^{N-1} |x_2 - x_1| < \epsilon \cdot (1 - \beta)$

Then for any  $m \geq n \geq N$ ,

$$\begin{aligned} |x_m - x_n| &\leq |x_m - x_{m-1}| + |x_{m-1} - x_{m-2}| + \dots + |x_{n+1} - x_n| \\ &\leq (\beta^{m-2} + \beta^{m-1} + \dots + \beta^{n-1}) \cdot |x_2 - x_1| \\ &= \beta^{n-1} (1 + \beta + \dots + \beta^{m-n-1}) \cdot |x_2 - x_1| \\ &< \beta^{n-1} \left( \sum_{k=0}^{\infty} \beta^k \right) \cdot |x_2 - x_1| = \beta^{n-1} \cdot \frac{1}{1-\beta} \cdot |x_2 - x_1| < \epsilon. \end{aligned}$$

Therefore  $\{x_n\}$  is Cauchy sequence in  $\mathbb{R}$ , the complete metric space, thus

$$x^* = \lim_{n \rightarrow \infty} x_n$$

exists. Finally, the differentiable map continuous,

$$f(x^*) = f\left(\lim_{n \rightarrow \infty} x_n\right) = \lim_{n \rightarrow \infty} f(x_n) = \lim_{n \rightarrow \infty} x_{n+1} = x^*.$$

□

Note: 수렴사상  $\rightarrow$  MVT  $\rightarrow$  difference of Adjacent 가 작아짐으로 같음  $\rightarrow$  근사하면  $\rightarrow$  연속함수  $\rightarrow$  연속수렴보존

⌈ If a continuous map  $f: \mathbb{R} \rightarrow \mathbb{R}$  satisfies:  $f$  has a local minimum at for any  $x \in \mathbb{R}$ , then  $f$  is constant.

Proof. Let  $a \in \mathbb{R}$  be given. Consider a set

$$A \stackrel{\text{def}}{=} \{x \in \mathbb{R} \mid f(x) \leq f(a)\} = f^{-1}([f(a), \infty))$$

Since  $f$  is continuous,  $A$  is closed set. Meanwhile, for any  $b \in A$ ,  $b$  is a local minimum:

there exists an open  $U$  containing  $b$  s.t.  $\forall x \in U$ ,  $f(b) \leq f(x)$

But  $f(a) \leq f(b)$ , thus  $U \subseteq A$ . Now,  $A$  is non-empty clopen, thus  $A = \mathbb{R}$ .

Finally, for any  $a, b \in \mathbb{R}$ ,  $f(a) \leq f(b)$  and  $f(a) \geq f(b)$

Section 8.23.

Define  $f(x) = \frac{x^3 + 1}{3}$ . It has three fixed points  $\alpha, \beta, \gamma$  where  
 $-2 < \alpha < -1, 0 < \beta < 1, 1 < \gamma < 2$

For arbitrary  $x_1 \in \mathbb{R}$ , define  $\{x_n\}$  as  $x_{n+1} = f(x_n)$ ,  $n \geq 1$ .

1) If  $x_1 < \alpha$ , then  $x_n \rightarrow -\infty$  as  $n \rightarrow \infty$

2) If  $\alpha < x_1 < \gamma$ , then  $x_n \rightarrow \beta$  as  $n \rightarrow \infty$

3) If  $\gamma < x_1$ , then  $x_n \rightarrow \infty$  as  $n \rightarrow \infty$

Proof. Observe that  $f'(x) = x^2$ . And, for each  $n \geq 1$ ,

$$x_{n+1} - x_n = f(x_n) - x_n = \frac{x_n^3 + 1}{3} - x_n$$

Suppose that  $x_1 < \alpha$ .

If  $x_n < \alpha$ , then  $x_{n+1} - x_n = f(x_n) - x_n < 0$ , thus  $x_{n+1} < x_n < \alpha$ .

Now, for each  $n \geq 1$ ,  $x_{n+1} - x_n = f(x_n) - f(x_{n-1}) = f'(t_n)(x_n - x_{n-1})$  for some  $t_n < \alpha < -1$ .

For  $n=1$ ,  $x_2 - x_1 = f(x_1) - x_1 < 0$ .

And, if  $x_n - x_{n-1} < 0$ , then  $x_{n+1} - x_n = f(x_n) - x_n$

Then for each  $n \geq 1$ ,

$$x_{n+1} - x_n = f(x_n) - f(x_{n-1}) = f'(t_n)(x_n - x_{n-1}) \quad \text{for some}$$



